

Education

Sun Yat-sen University
GPA: 3.95/4.0 (Rank: 4/166, Top 2.5%)
Selected Courses: Operating System (98), Machine Learning (96), Introduction to Robotics (95), Advanced Programming (95), Linear Algebra (95)

Sep 2023 – Expected June 2027
Bachelor of Engineering in Artificial Intelligence

Research Experience

Research Assistant: Lifelong In-Context Reinforcement Learning
Advisors: Prof. Qin Jin & Ye Wang (Ph.D. Candidate)

Renmin University of China
July 2025 – Feb 2026

- **Paradigm Proposal:** Proposed and formalized the "Lifelong ICRL" paradigm to address the continuous adaptation challenge of agents facing stochastic mutations and structured distribution shifts within a single lifecycle.
- **Framework Development:** Developed a training and evaluation framework for non-stationary environments using JAX-based high-speed parallel simulators (XLand-Minigrid and Kinetix). Implemented three core testing protocols: completely stationary, stochastic mutation, and structured non-stationary (e.g., linear drift, periodic cycles).
- **Architecture Evaluation:** Evaluated Mamba2 (SSM), Gated DeltaNet (Linear Attention), and Hybrid attention architectures, focusing on the **"memory vs. forgetting"** dilemma in LLM agents. Demonstrated the synergistic advantages of local attention (precise backtracking) and recurrent state compression (global belief) in adapting to environmental evolution.

Submission: "Exploring Lifelong Adaptation: In-Context Reinforcement Learning in Non-Stationary Environments" **Co-first Author, ICML 2026 Under Review.**

Project Lead: Multi-Turn Image Editing
Advisor: Assoc. Prof. Xudong Mao, National Innovation Training Program

School of Artificial Intelligence, SYSU
Sept 2024 – Nov 2025

- **Algorithm Implementation:** Proposed and implemented FreqEdit, a training-free frequency-domain feature preservation framework, to mitigate high-frequency detail loss and subject distortion in continuous multi-turn instruction-driven editing models (e.g., FLUX.1 Kontext, Qwen-Image).
- **Benchmark Construction:** Constructed an automated multi-turn editing benchmark (70 image sets, 10 continuous complex instructions per image) using LLM-generated progressive prompts. Designed a multi-dimensional quantitative evaluation pipeline combining LPIPS, CLIP-I, DINO-v2, and GPT-4o/HPSv3 to reduce the reliance on manual qualitative assessments.

Validation & Cross-Architecture Adaptation: Validated the core components of FreqEdit (adaptive feature injection and path compensation) and engineered their adaptation across mainstream Flow Matching architectures (e.g., FLUX.1 Kontext, Qwen-Image). Conducted ablation studies to confirm the framework's effectiveness against texture collapse and noise accumulation, demonstrating improvements in subject consistency and instruction adherence over existing open-source baselines.

Submission: "FreqEdit: Preserving High-Frequency Features for Robust Multi-Turn Image Editing" **(Co-first Author, CVPR 2026 Under Review)** [[arXiv:2512.01755](#)].

Selected Awards

National Scholarship	Highest Honor for undergraduates in China	2024, 2025
Baosteel Excellent Student Scholarship	Awarded to only 504 students nationwide	2025
SYSU Outstanding Scholarship (First Class)	Top 5% in School of AI	2024, 2025

Selected Projects

Training-Free Cross-Modal In-Context Learning via Signal Processing
Processing

Course Project: Digital Signal

Proposed a novel cross-modal alignment paradigm by conceptualizing 3D molecular embeddings as discrete digital signals. Applied Graph Laplacian reordering and Discrete Cosine Transform (DCT) to extract and preserve high-frequency features, enabling LLMs to directly interpret cross-modal frequency-domain fingerprints without explicit training.

NoteAgent: AI-Driven Interactive Teaching Assistant

Course Project: Human-Computer Interaction

Developed a full-stack educational platform using FastAPI and React, integrating a **Retrieval-Augmented Generation (RAG)** question-answering system. Implemented bidirectional timestamp anchoring to synchronize video content with notes, effectively reducing cognitive friction and enhancing knowledge internalization in online learning environments.

Technical Skills

Frameworks& Tools

Python (PyTorch, JAX), C/C++, LaTeX, Linux (Shell), Git, Docker

Languages

English (IELTS: Preparing, Expected: 2026.05;CET6)

Leadership & Service

President

School of Artificial Intelligence, SYSU

Student Union

Nov 2024 – Nov 2025

Founded and open-sourced **AICompass**, a comprehensive academic repository sharing structured learning roadmaps, course materials, and review guides. Concurrently directed the **Peer Tutoring** program, fostering a collaborative community among 200+ undergraduate peers.

Led an executive team to organize large-scale campus initiatives (including athletic tournaments and cultural festivals), demonstrating strong capabilities in project management, resource allocation, and cross-functional communication.